

Advanced Computer Algorithms
(ET6501701)
Term Projects

Date: September 3, 2025

Instructor: M. B. Lin

Due: December 10, 2025

Term Projects: In this semester we have ten project options. You only need to pick one of them and finish it completely and hand it out on schedule. You may be referred to any resource so long as cite it but are needed to verify your resulting design (IP) as thoroughly as you can. In addition, assume that the resulting design will be housed in a 40-pin package, in which 8 pins are reserved for V_{DD} and V_{SS} . Of course, **any academic dishonesty is strongly prohibited**. Your work will be evaluated in terms of *performance*, *power dissipation*, and *hardware cost* in addition to the verification of correctness.

1. **A Brent-Kung parallel-prefix adder with at least 256-bit inputs**
2. **A Kogge-Stone parallel-prefix adder with at least 256-bit inputs**
3. **A Ladner-Fisher parallel-prefix adder with at least 256-bit inputs**
4. **A carry-skip adder with at least 256-bit inputs**
5. **A sorting network with at least 16 inputs with at least 8-bit each**
6. **A 64×64 -bit or bigger signed multiplier with an unsigned 8×8 array multiplier**
7. **A FFT network with at least 32 points with each input at least 16 bits**
8. **An AES encryption/decryption module with a key size of at least 128 bits**
9. **A RSA encryption/decryption module with with a key size of at least 128 bits**
10. **A (15, 11) Reed-Solomon decoder with a generator $G(x)$ of $x^4 + 15x^3 + 3x^2 + x + 12$**